

URL shortner

Functional requirements:

user should be able to shorten long URL based on his preference,
all the shortened URL should be unique ,

Non-functional requirements:

data-partitioning,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability,
low latency,
high throughput,
CAP theorem (Here Eventual consistency is fine , but here we need 99.99% uptime i.e
Availability>>Consistency)

Core Entities:

user,
url,
history

API :

POST:/shorten

```
{  
  userID,  
  {LongURL}  
}
```

GET:/click/{url}

HLD :

